

Problem-Solution Fit

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] i. Residents living in flood-prone areas ii. Disaster Management Authorities iii. Farmers iv. Local Government Agencies v. Environmental Researchers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited internet access in some regions. - Lack of technical knowledge. - No access to expensive forecasting systems. - Dependence on publicly available weather information.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions - Government weather alerts - Manual weather monitoring - News broadcasts Pros - Easily accessible - Official information Cons - Generalized information - Delayed updates - Limited prediction capability
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty predicting floods in advance. - Heavy rainfall causes unexpected flooding. - Lack of quick flood	9. PROBLEM ROOT / CAUSE [RC] i. Unpredictable weather conditions. ii. Heavy seasonal rainfall.	7. BEHAVIOR + ITS INTENSITY [BE] i. Regularly checks weather updates during rainy seasons. ii. Takes precautionary

warning systems. Frequent during monsoon and heavy rainfall seasons.	iii. Climate change. iv. Lack of accessible real-time flood prediction systems.	measures when flood warnings are issued. iii. High activity during heavy rainfall periods.
3. TRIGGERS TO ACT [TR] i. Heavy rainfall forecasts. ii. Rising river water levels. iii. Sudden weather changes. iv. Need for early flood preparedness.	10. YOUR SOLUTION [SL] Rising Waters is a Machine Learning-based flood prediction web application that analyzes weather parameters such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, and average rainfall using a Random Forest Classifier. The system provides instant predictions indicating whether there is a Flood Chance or No Flood Chance, enabling users to take preventive measures and improve disaster preparedness.	8. CHANNELS OF BEHAVIOR [CH] ONLINE i. Flood Prediction Web Application ii. Weather websites iii. Government portals
4. EMOTIONS BEFORE / AFTER [EM] Before - Worried - Uncertain - Anxious After - Informed - Prepared - Confident		OFFLINE - Television - Radio - Local disaster management announcements